Chapters from the field of psychological methodologies

Introduction to model-based statistical methods, linear regression and it's use for mediation and moderation analysis

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The goal of the course is to give an overview of central terms used when working with model-based statistical methods, like Linear Regression Analysis, Logistic Regression Analysis, Structural Equation Modeling etc., and to give an introduction to examining mediation and moderation effects using Multiple Linear Regression Analysis.

The course duration is 10 hours, it is broken up into six sessions, approximately 80-90 minutes each. We use the jamovi statistical software package, and the additional modules *jamm*, *medmod*, and *GLMj3* in the examples (introductory session for jamovi included).

Suggested readings and other sources

Hayes, A. F. (2022). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach.* Guilford Press.

The jamovi project (2025). *jamovi (Version 2.7) [Computer Software]. [Available from url:* https://jamovi.org]

Navarro, D. J., Foxcroft, D. R. (2025). *Learning statistics with jamovi: A tutorial for beginners in statistical analysis.* Open Book Publishers. [Available from url: https://www.learnstatswithjamovi.com]

Overview of sessions

Day 1

Session 1: Theoretical foundations

Session 2: Short introduction to jamovi

Session 3: Simple and Multiple Linear Regression Analysis (in jamovi)

Day 2

Session 4: Moderation analysis in jamovi (using *medmod* and *GLMj3*)

Session 5: Mediation analysis in jamovi (using *medmod* and *jamm*)

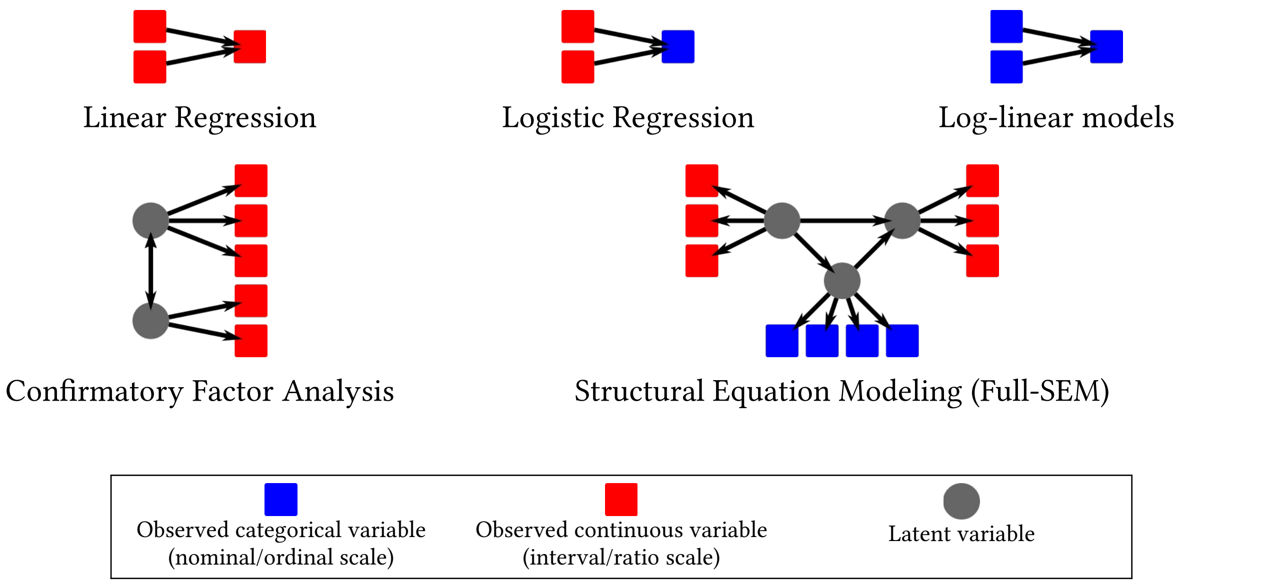
Session 6: Advanced mediation models (mediation with moderation) in jamovi (using *jamm*)

Session 1
Theoretical foundations

Introduction

- Statistical modeling is a general purpose approach to study relationships between variables
- Statistical models are:
  - Abstract and formal hypotheses about the “data-generating” process
  - A set of variables, a specific structure (relationships between variables) plus noise (measurement error, and omitted variables)
  - A simplified instead of a thorough description of reality
- Models can be used to:
  - Explain: understand relationships between variables structurally
  - Predict: estimate unknown outcomes
- Models are versatile – even some classical statistical methods like ANOVA, ANCOVA or even independent samples t-test can be thought of as special cases of a statistical model (the linear model, specifically)

Examples of various statistical models



- Some well known and widely used models:
  - General Linear Model
    - special cases: Simple Linear Regression, Multiple Linear Regression, Multivariate Linear Regression, ANOVA, ANCOVA, MANOVA, independent samples t-test
  - Generalized Linear Model

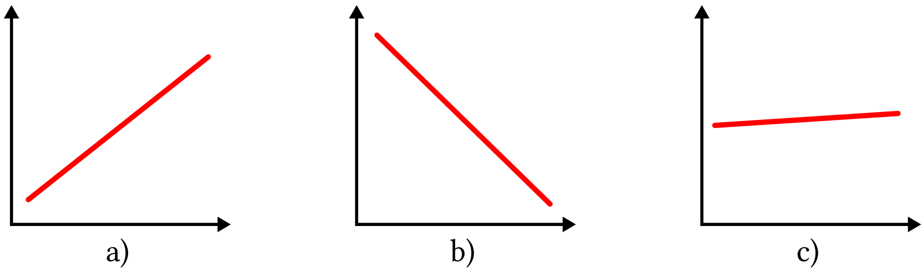
    - special cases: Binomial Logistic Regression, Ordinal Regression, General Linear Model (!)
  - Structural Equation Modeling
    - special cases: Path Analysis, Confirmatory Factor Analysis, Full SEM
- Variables in a model:
  - independent (predictor) and dependent (target) variables (regression)
  - latent variables, and observed variables (SEM)
  - factors, observed variables (factor analyses)

Steps and typical activities of model-based statistical analyses

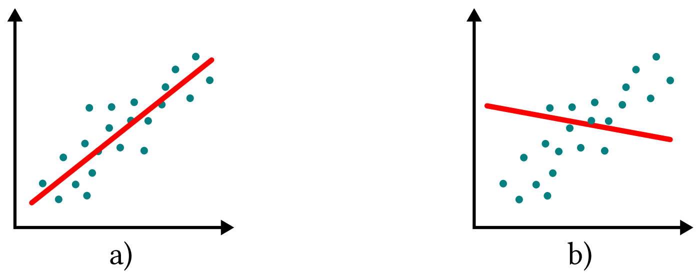
Parameter estimation

- Models have **parameters** besides variables – they make them adaptable, by defining how they “behave”

Example: the linear model: $Y = b_0 + b_1 X + \epsilon$



- Parameters are unknown quantities – we have to find the parameters that are best at describing the distribution of data



- **Parameter estimation** is the process of using observed data to determine the values of unknown values (parameters) in a model
  - various estimation techniques: eg. Maximum Likelihood Estimation (MLE), Ordinary Least Squares (OLS)
    - each estimation technique has some **assumptions** – if they are violated, estimations can be biased, or uncertain
- **Non-identifiability**: when parameters can not be uniquely determined for a model
  - too many parameters for too little data

- latent variable models (more unknowns than information)

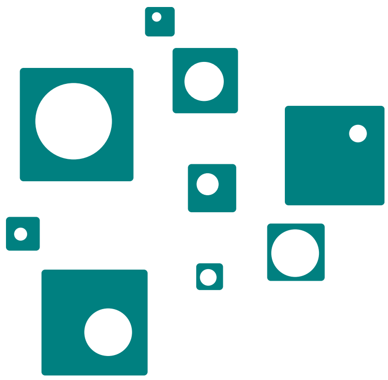
- Amount of data will affect **uncertainty in parameter estimates** (less data, more uncertainty)

Assessing model fit

- **Model fit** describes
  - how well the model matches the observed data, or
  - how closely **predictions** align with the data used to estimate parameters
- Various ways to measure model fit:
  - size of residuals (prediction errors)
  - likelihood-based fit measures (likelihood of data given the model)
  - proportion of explained variance

Why do some squares have smaller areas and others larger areas?
(proportion of explained variance)

side length alone explains the entire variance



side length alone explains only part of the variance

side length and *diameter of hole* explains the entire variance

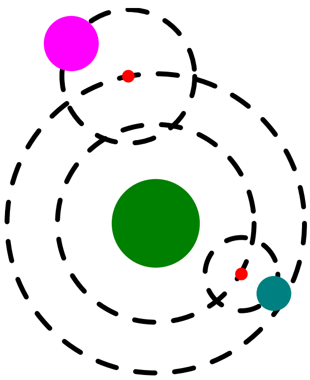
Inference about model parameters

- Conclusions about the underlying model structure (eg. relationships/effects in the model)
- Questions like:
  - Size of effects (point estimate)

    - Direction of effects
    - Uncertainty in estimate (standard error, confidence interval)
    - Is the parameter statistically different from zero (hypothesis tests/p-values)

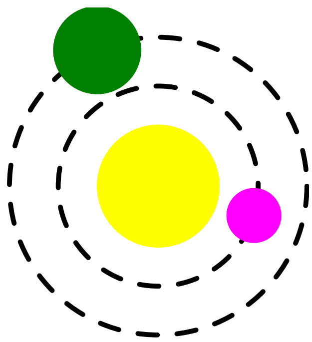
Comparing models

- Out of two models (a simpler and a more complex one) for the same data, which one should we prefer?
- A more complex model always fits the data better (explanatory power)
- Parsimony: simpler models are preferred to more complex models (Occam's razor)
- Solution: trade-off between explanatory power and complexity
    - A more complex model is preferred over a simpler model only when it has a significantly greater explanatory power (fits the data significantly better)



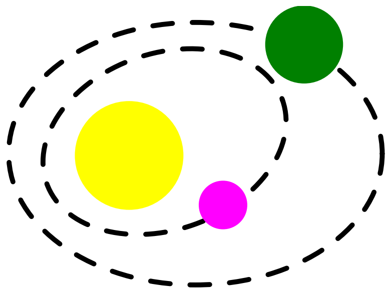
Ptolemy
Earth in the center
Planets move on *epicycles*
Epicycles move around *deferents*

Model complexity: large
Predictions: very accurate



Copernicus
Sun in the center
Planets move on circles around the Sun

Model complexity: low
Predictions: less accurate



Kepler
Sun in the center
Planets move on elliptic paths around the Sun

Model complexity: low
Predictions: very accurate

Session 2
Short introduction to jamovi

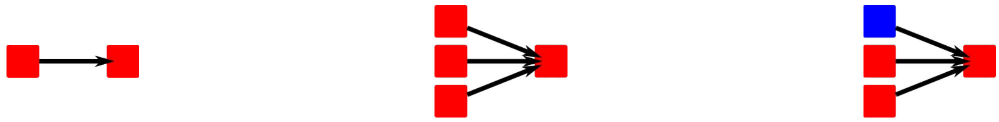
Some advantages of jamovi

- Free and open-source
- User-friendly interface (no coding required)
- Live output
- Extensible (plugin system)
- R included

Beginner-level jamovi

- The interface
- Reading and importing data files (csv files)
- Setting variable attributes (name, description, measure type)
- Inspecting data: descriptive statistics
- Simple analyses (eg. t-tests, correlation analysis)

Session 3
Simple and Multiple Linear Regression Analysis (in jamovi)

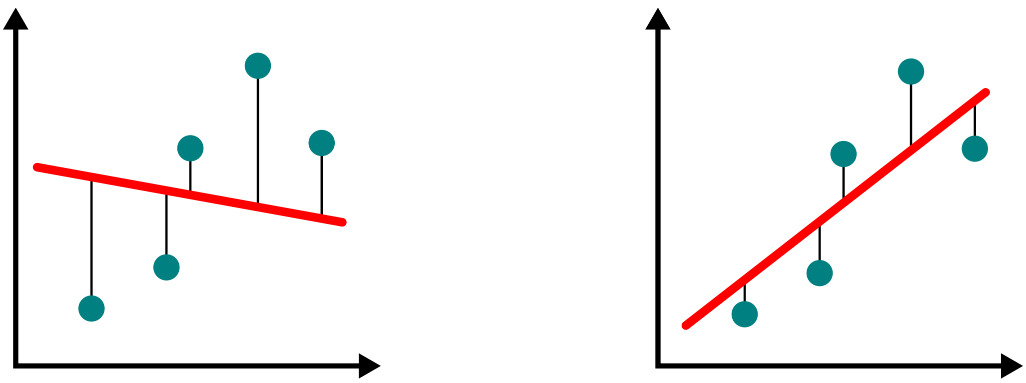


Reviewing the linear model

$$Y = b_0 + b_1 X_1 + b_2 X_2 + \ldots + b_k X_k + \epsilon$$

Y	*dependent* (target) variable
X_1 - X_k	arbitrary number of *independent* (predictor) variables
b_0	*intercept* parameter
b_1-b_k	*slope* parameters
ε	*random error* term

- Parameter estimation: Ordinary Least Squares (OLS)
  - parameters with the smallest *residuals* (prediction errors)



a) b)

Example: acceptance of self-driving cars

Assessing model fit

- Coefficient of determination, R^2, adjusted R^2
- Overall model fit: F-test

Inference

- parameter estimates, standardized parameter estimates
- one-sample t-test

Model comparisons

- Hierarchical Linear Regression Analysis
- Change in model fit: ΔR^2, F-test

Checking assumptions

- Normality of residuals
- Collinearity

Using categorical variables as independent variables

- special coding of variables (dummy coding)
- interpreting effects for categorical variables